

CHAPTER 1

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

GLOBE AND MAP SKILLS**Learning Outcomes**

After studying this chapter the students will be able to:

1. Understand that there are 180 imaginary lines of latitude and 360 imaginary lines of longitudes.
2. Name the main lines of latitudes and longitudes.
3. Locate the main lines of latitudes and longitudes on a globe and on a map.
4. Use latitudes and longitudes to locate major cities of Pakistan and of the world.
5. Use the index of an atlas to locate places.
6. Use latitudes and longitudes in determining directions.
7. Identify time zones and relate them to longitudes.
8. Identify the significance of the location of Pakistan.
9. Recognize that there are many kinds of maps and choose the best map for the purpose at hand.
10. Read and interpret scales (expressed as a statement or bar) on different maps.
11. Use the map scale to measure roads and rivers, and determine distance between places.

GLOBE AND MAP SKILLS

THE LOCATION OF PAKISTAN IN RELATION TO HER NEIGHBOURS

Look at the map of the world in figure 1.1. There are seven large land masses called continents. Find the largest continent. It is Asia. Pakistan is situated in the southern part of Asia, commonly known as South Asia.



Fig 1.1 Map of the World

Pakistan shares borders with four countries: Iran, Afghanistan, China and India. Iran lies to the west of Pakistan. To the north-west is Afghanistan. China lies to the north-east of Pakistan. Pakistan shares only a small border with China which lies in the Karakoram Mountains. India lies to the east of Pakistan. The Arabian Sea lies in the south of Pakistan.

We have just learnt the location of Pakistan in relation to her neighbours. We can also find the exact location of Pakistan using the lines of latitude and longitude. Let us first learn what lines of latitude and longitude are.



Fig 1.2 Map showing the neighbouring countries of Pakistan

LINES OF LATITUDE AND LONGITUDE

To help us locate places on globes and maps, grid lines are used . These lines divide the globe and map into grid squares (see fig. 1.3). The vertical grid lines  are called lines of longitude and are marked with 'numbers'. The horizontal grid lines  are called lines of latitude and are marked with 'letters'.



Fig 1.3 Map Showing the Vertical and Horizontal Grid Lines



Fig 1.4 World map showing the grid formed by the lines of latitude and longitude

Lines of Latitude or Parallels of Latitude

The Earth is not a perfect sphere; it is slightly flattened at the North Pole and the South Pole. Exactly half way between the North Pole and the South Pole is the most important line of latitude, the Equator. The Equator divides the Earth into two equal halves or hemispheres (half spheres): the half which is located to the north of the Equator, is called the Northern Hemisphere and half which is located in the South of the Equator is called the Southern Hemisphere (see fig. 1.5).

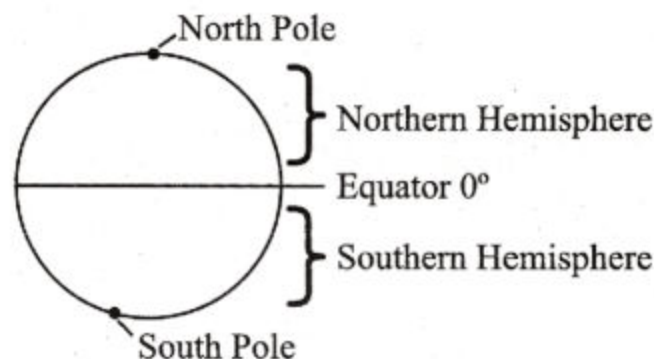
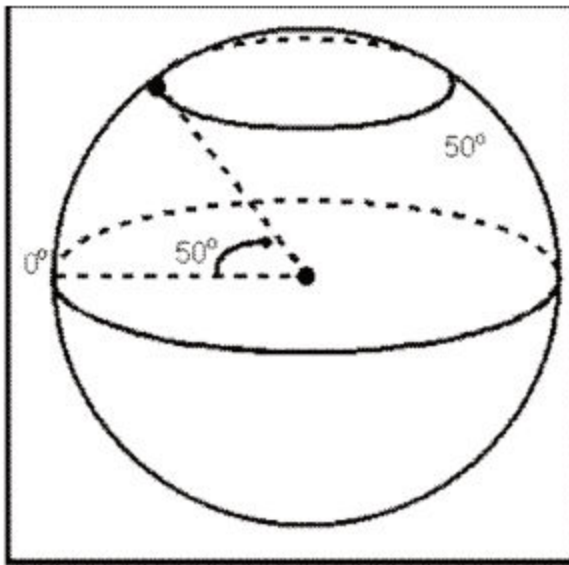
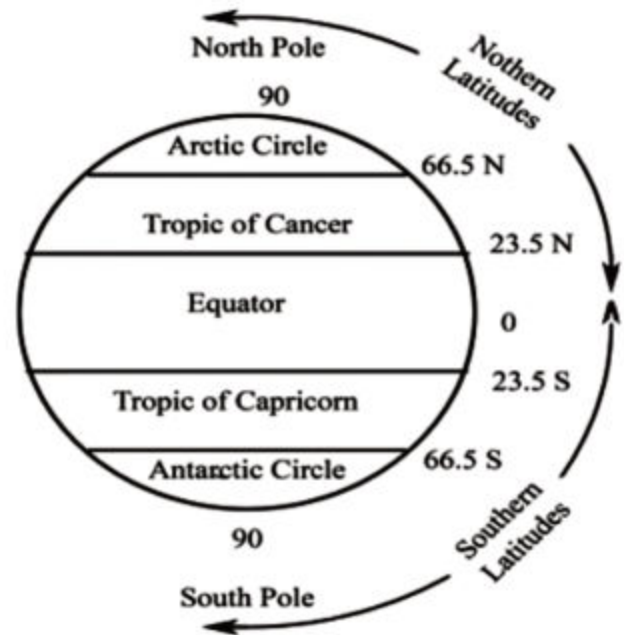


Fig. 1.5 The Northern and Southern Hemispheres

There are 90 imaginary lines of latitude in each hemisphere. That is, 90 equally distanced parallel lines north of the Equator and 90 equally distanced parallel lines south of the Equator. The lines of latitude run in the east-west direction in complete circles around the Earth. The bold lines show the lines of latitude that are important enough to be given special names (see fig 1.6).



The method of identifying lines of latitude



The important lines of Latitudes

Fig 1.6 Lines of the Latitude**i. The Equator**

The 0° latitude is called the Equator, because it divides the Earth into two equal halves: the Northern Hemisphere and the Southern Hemisphere.

ii. The Tropics of Cancer and Capricorn

There are two tropics. The latitude located at 23.5° north of the Equator is called the Tropic of Cancer. The latitude located at 23.5° south of the Equator is called the Tropic of Capricorn.

iii. The Arctic and Antarctic Circles

The latitude located at 66.5° north of the Equator is called the Arctic Circle. The latitude located at 66.5° south of the Equator is called the Antarctic Circle.

iv. The North and South Poles

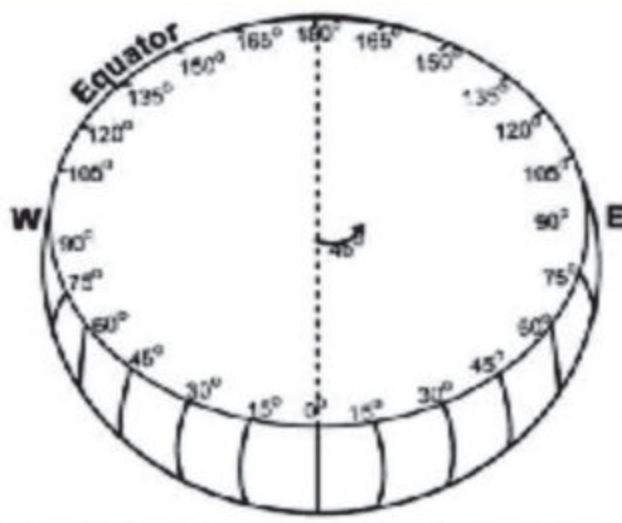
The Pole located at 90° north of the Equator is called North Pole. The Pole located at 90° south of the Equator is called South Pole.

When you give the latitude of a place, you must always say whether it is north or south of the Equator. For example, the Arctic Circle is at 66.5° N while the Antarctic Circle is at 66.5° S. The only exception is the Equator which lies at 0° .

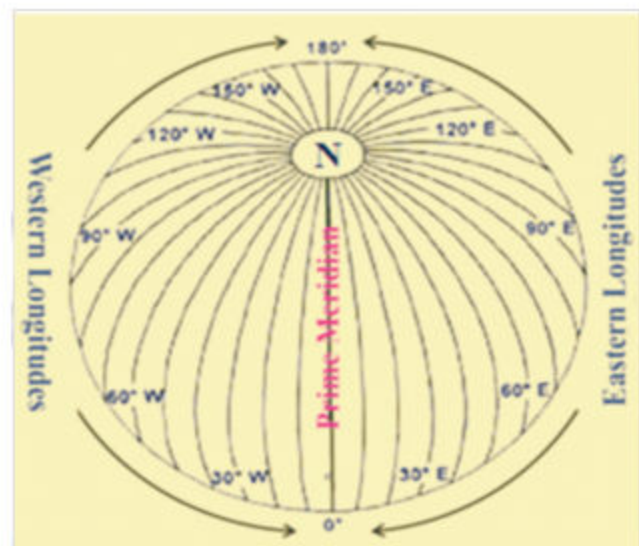
Lines of Longitude

Lines of longitude are imaginary lines drawn from the North Pole to the South Pole. That is, they run in the north-south direction. While each line of latitude completely encircles the Earth, each line of longitude goes only half way around the Earth.

The most important line of longitude is the Prime Meridian at 0. Any line of longitude could be the Prime Meridian but by international agreement the meridian that passes through the town of Greenwich, England is the official Prime Meridian. There are 180 lines of longitude east of the Prime Meridian which are marked with E and 180 lines of the longitude to the West which are marked with W (see fig 1.7). Because the Earth is sphere. There are 360 lines of longitude. 180 lines of longitude to the East and 180 lines of longitude to the west of the Prime Meridian.



The method of identifying lines of longitude



The prime meridian and longitude 180

Fig 1.7 Lines of Longitude

Two lines of longitude are very important and are given special names:

i. Prime Meridian

The most important line of longitude is the 0° longitude also called the Prime Meridian. It is important because it is the starting point for measuring distance between east and west around the Earth. It is also important as the Coordinated Universal Time by which all countries and regions measure their time is based on it.

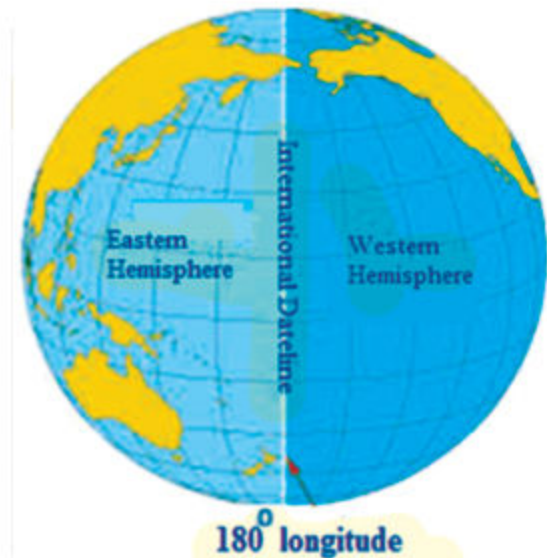
ii. International Dateline

Exactly opposite the 0° longitude or the Prime Meridian is the 180° longitude which is also known as the International Date Line (IDL). Countries and regions lying on the east and west of the IDL have different dates. As you pass the International Date Line, you either gain a day (going west) or lose a day (going east.). Look at figure 1.8 you will see that unlike all the other lines of longitude the IDL is not a straight

line. This is to allow an entire country or region to have the same date.



Fig 1.8 The International Date Line



The Prime Meridian and the IDL create a circle that divides the Earth into the eastern and western hemispheres. This is similar to the way the Equator at 0° latitude divides the Earth into the northern and southern hemispheres.

The eastern hemisphere is east of the Prime Meridian and west of the IDL. Asia, Australia, and most of Africa, are part of the eastern hemisphere. The western hemisphere is west of the Prime Meridian and east of the IDL. North and South America, the western part of the British Isles (including Ireland and Wales), and the northwestern part of Africa are part of the western hemisphere. Which hemisphere does Pakistan lie in?

Activity

Draw horizontal and vertical lines on a page in your book. Name the centre most horizontal line the Equator and mark off the others 10°N , 20°N as shown here. Name the centre most vertical line the Prime Meridian and mark off the other 10°E , 10°W . Draw a star in any grid and let your partner tell you which grid it is located in.



Draw a globe show the lines of longitude and latitude. Mark the 180° longitude as the IDL.

Identifying the exact location of a place using lines of latitude and longitude

If we look at the globe or a map, we will clearly see the lines of latitude and longitude. With the help of these lines, we can easily identify the exact location of a place on a map or globe. Let us learn how to identify the exact location of Islamabad by following the steps given below.



Fig 1.9 Map of Pakistan showing lines of Latitude and Longitude

1. Look for Islamabad on the map. It will be marked by a symbol that looks like this  .
2. Draw a horizontal line going through the symbol that marks Islamabad on the map. Divide the space between 32° N and 36° N into 4 equal parts. The horizontal line which you have drawn is very close to 33° N. The latitude is 33.42° N.
3. Draw a vertical line going through the symbol that marks Islamabad on the map. Divide the space between 72° E and 76° E into 4 equal parts. The vertical line which you have drawn is very close to 73° E. The longitude is 73.09° E
4. Read the latitude first. Read the longitude next. Therefore, Islamabad is at 33.42° N and 73.09° E.

Finding the exact location of place on map using lines of latitude and longitude

Every place shown in the atlas is listed in alphabetical order in the index at the back of the atlas. We will find entries like this Karachi 24.52°N 67.03°E . This tells us that Karachi is located at 24.52°N and 67.03°E . On the map in figure 1.10 find 24°N and draw a horizontal line very near it. Find 67°E and draw a vertical line very near it. You will see that Karachi is located exactly where the two lines meet.

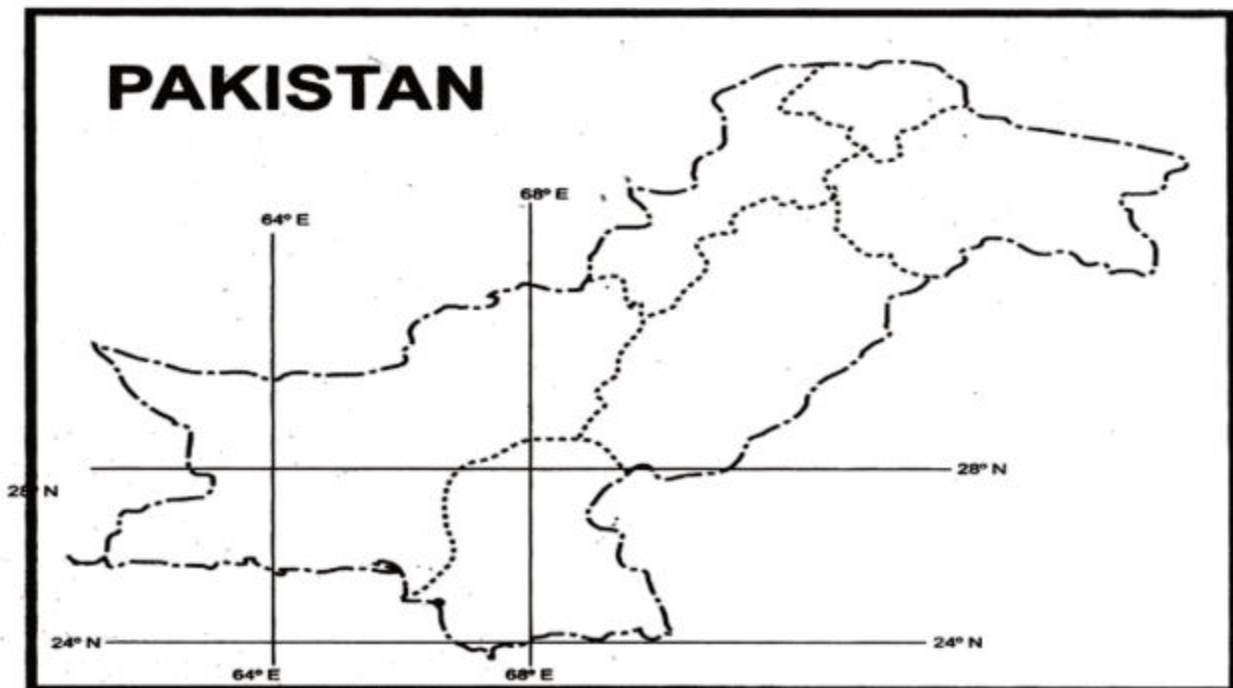


Fig. 1.10 Map of Pakistan

Activity

- Using the same process you have just learnt identify the exact location of Peshawar and Hyderabad.
- Find the cities located at i. 31.25°N 73.9°E ii. 25.22°N 68.22°E

The exact location of Pakistan

We have learnt how to identify the exact location of a place using the lines of latitude and longitude. To identify the exact location of a country, we have to find the furthestmost point of that country in each of the cardinal directions: north, south, east and west.

Let us identify the exact location of Pakistan (see fig. 1.10)

- Find the northern-most point of Pakistan and read the latitude. It is 36.75°N .
- Find the southern-most point and read the latitude. It is 23.45°N .
- Find its eastern-most point and read the longitude. It is 75.5°E .
- Find the western-most point and read the longitude. It is 61°E .

Thus we can say that Pakistan is situated in the Northern Hemisphere between 23.45° N and 36.75° N and between 61° E and 75.5° E.

Importance of Pakistan's Location

As we have just learnt, Pakistan is located in the Northern hemisphere between the latitudes of 25° N and 37° N. Since this region experiences all the four seasons in a year, a great variety of crops are grown and livestock raised here.

We have also learnt that Pakistan is located in the continent of Asia, in the region known as South Asia. Because of this it is a member of the South Asian Association for Regional Cooperation (SAARC). It has a very friendly relationship with the highly industrialized country of China in the North. There are many Muslim countries that are situated close to it and with which it has friendly ties. The Arabian Sea, which is located in the south of Pakistan, plays an important role in international trade.

Scale

It is impossible to draw a map the same size of the feature on the ground. A map, therefore, is a scaled down version of the feature on the ground. A map scale compares the distances on a map to the actual distance on the Earth's surface. The scale of a map can be expressed in three ways: as a statement, a line or a ratio.

i. Statement Scale

The scale that is described in words and numbers as shown in figure 1.10 is called a statement scale. The given scale means 1 centimeter (cm) on the map is equal to 10 kilometers (kms) on the ground..

1 cm on the map represents 10 kms on the ground

Fig 1.11 A Scale statement

ii. Line Scale

In the line scale, a line several centimeters long is divided into equal parts. In figure 1.11 the distance between 0 and 100 kilometer is 2 centimeters. This means that 2 centimeters on the map is equal to 100 kilometers on the ground or 1 centimeter on the map is equal to 50 kilometers on the ground.

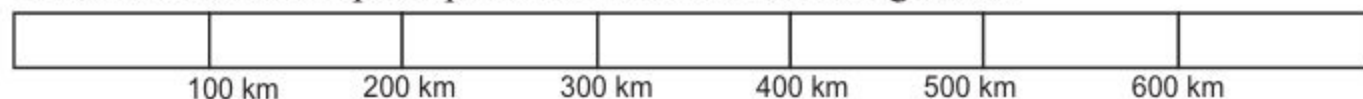
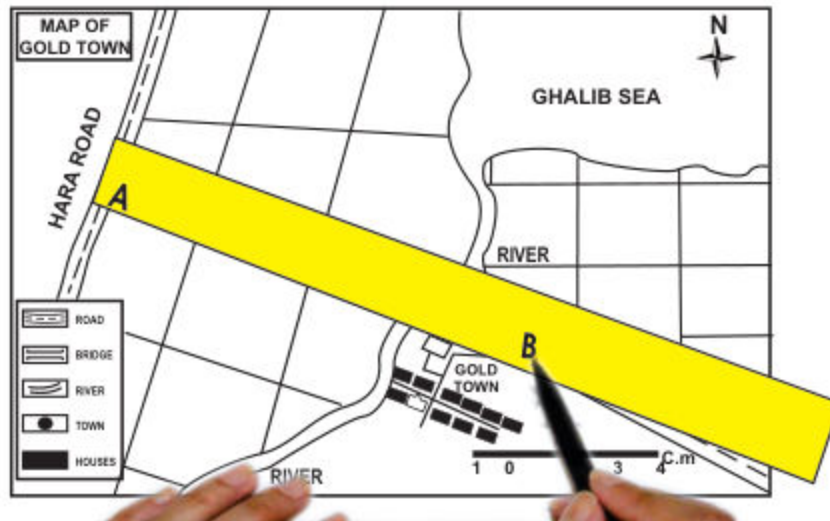


Fig 1.12 A Line scale

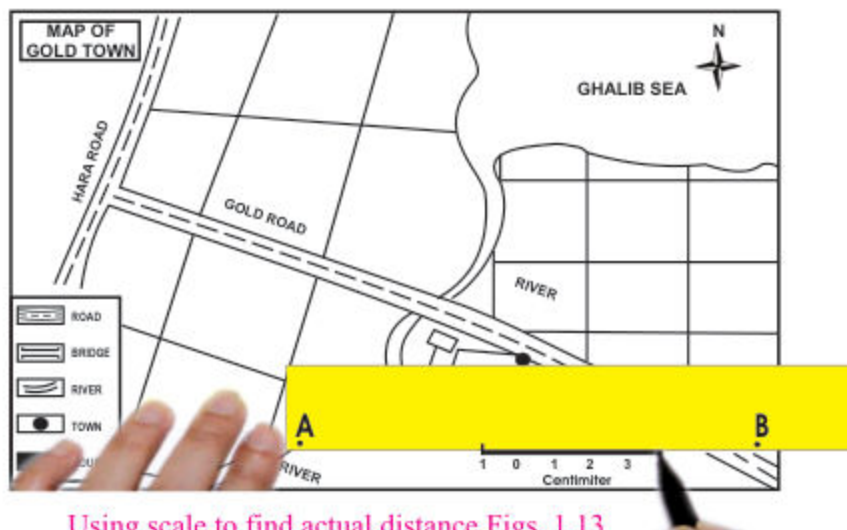
iii. Ratio or Fractional Scale

In this type of scale, no unit of distance is mentioned. Rather, a ratio is established that can be put into all units of distance. For instance, 1: 100. In this

ratio, if 1 means 1 centimeter, 100 would mean 100 centimeters and if 1 means 1 kilometer, 100 would mean 100 kilometers.



A child measuring the distance between Hara road and Gold town.



Using scale to find actual distance Figs. 1.13

Let us learn how to measure the distance between two places by measuring the distance between Hara Road and Gold Town.

1. Look for Hara Road and Gold Town on the map
2. Place a straight piece of paper between Hara Road and Gold Town. Mark Hara Road as A and Gold Town as B on the paper.
3. Place the paper against the line scale on the map with A on the 0 mark. note the position of B. In this case, the distance is more than 13 kilometers.
4. To find out exactly how much over 13 kilometers the distance is, you can place the paper on the scale to the left of the mark and identify the fraction.
5. The distance between Hara Road and Gold Town is 13.5 kilometers.

Activity

Using the map in figure. 1.9 measure the distance between Karachi and Islamabad. What is the distance on the ground?

Using scale to measure roads and rivers

Roads and rivers are not usually straight, they are often curved. To measure curved distances, you should use a piece of thread.

1. Put one end of the thread at one end of the road or river to be measured.
2. Trace the length of the road or river with the thread.
3. When you come to the end of the road or river mark the thread.
4. Now put the thread on the scale and measure it as you did to identify distance Between two places.
5. Write down the reading.

Activity

Using a piece of thread, measure the length of the River Indus on a map of Pakistan. How long is it.

World Time- Zones

The world time-zones are based on the fact that the Earth moves through 15 degree of longitude each hour. Therefore, there are 24 standard time-zones ($24 \text{ hours} \times 15^\circ = 360^\circ$) (see fig. 1.14). Time zones are counted from the Prime Meridian (0° longitude), which runs through Greenwich, England. Each time zone is based on a central meridian, counted at 15° intervals from the Prime Meridian, and extends $7\frac{1}{2}^\circ$ to either side of the central meridian. The first time-zone is marked as "0" which lies between 7.5° longitude on either side of the Prime Meridian. The last time-zone is marked as "12" which lies between 172.5° longitudes on either side of the International Date Line. The time-zones that lie to the east of the 0 time-zone are denoted with plus (+) sign, whereas the time-zones that lie to the west of the 0 time-zone are denoted with minus (-) sign. The 0 and 12th time-zones are common to both the eastern and western groups. With the 0 time zone, neither the plus nor the minus sign is used, whereas with the 12th time zone, both plus and minus signs are used.

The world time-zones establish a relationship between the lines of longitude and time. The difference of time between any two adjacent time-zones is one hour. It means, for every 15° of longitudes the difference of time is 1 hour (4 minutes for 1° of longitude). Pakistan is located in the +5 time-zone; therefore, its time is five hours earlier than the time of London which is located in the 0 time-zone. Thus, when it is 8pm (night) in Pakistan, it is 3pm (afternoon) in London on the same date.

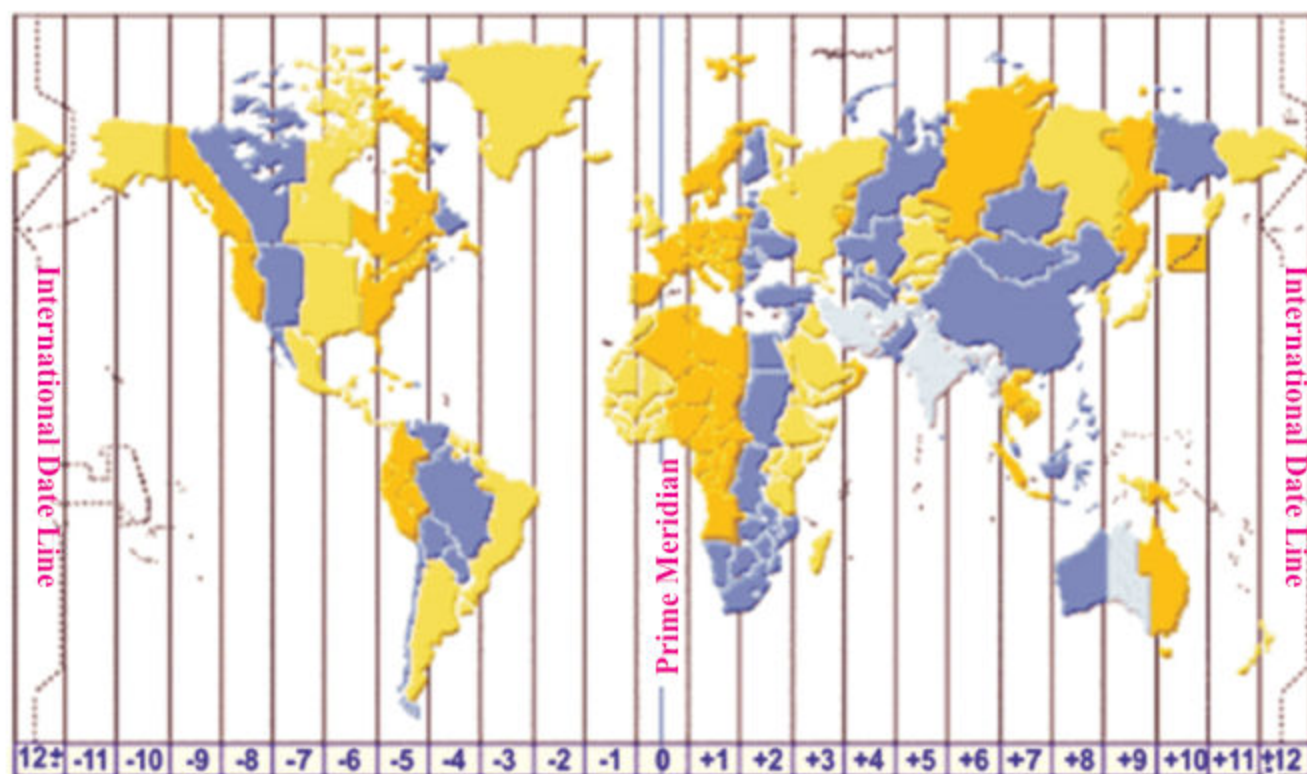


Fig. 1.14 World time zones

Kinds of Maps

A map is a drawing which shows the whole or a part of the Earth's surface. Maps show where one place is in relation to another. There are many kinds of maps. Three kinds of maps are explained below.

Physical Maps

Look at figure.2.1 on page 18. This is a physical map. It uses colour to show physical features such as mountains, hills, rivers and seas.

Political Maps

Look at figure 1.1 and 1.2. These are political or administrative maps. Political maps show the borders between countries and cities.

Relief Maps

Look at figure 2.3 on page 22. This is a relief map. It shows the height of hills and mountains and the depth of seas and oceans using different shades of color.

SUMMARY

In this chapter, we have learnt that there are 180 imaginary lines of latitude and 360 imaginary lines of longitude and that these lines help us locate places on maps and globes. The lines of longitude also help us determine the time in places around the world. We have also learnt that there are many different kinds of maps such as physical, political and relief maps and that all maps have scales. The scale on a map may be expressed as a statement, a bar or a ratio and we can use the map scale to determine distance between places as well as to measure the length of features such as rivers and roads.

EXERCISE

A. Multiple choice questions

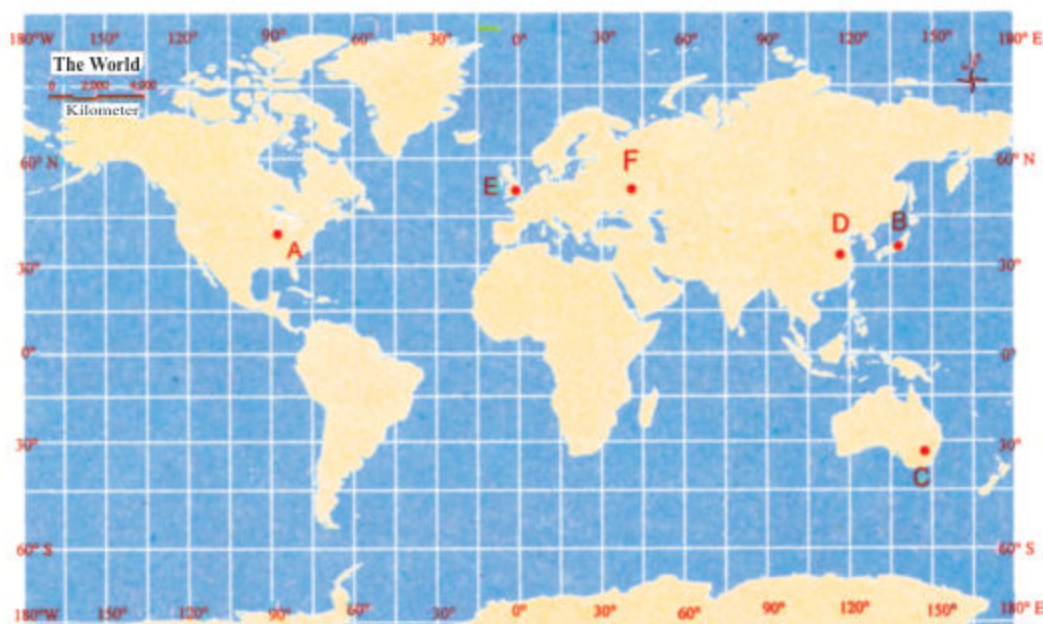
1. For every question, three answers are given; tick the correct one.
 - i. The imaginary lines that run in the north-south direction on a map are called
 - a. Latitudes
 - b. Longitudes
 - c. Parallels
 - ii. The latitude at 23.5 S is called
 - a. Arctic Circle
 - b. Tropic of Cancer
 - c. Tropic of Capricorn
 - iii. Into how many time-zones has the world been divided?
 - a. 12
 - b. 15
 - c. 24
 - iv. The map, that show us the border between countries are
 - a. Physical Maps
 - b. Political Maps
 - c. Relief Maps
 - v. What map would show us mountains and rivers
 - a. Physical Maps
 - b. Political Maps
 - c. Relief Maps

B. Answer the following questions

1. How many imaginary lines of latitude and longitude are there?
2. Name the main lines of latitude and longitude.
3. Name and describe the types of map scales.
4. State three reasons why the location of Pakistan is important.

C. Complete the table

1. The figure below shows a map of the world.
 - a. Identify the main lines of latitude and longitude on the map
 - b. The capital cities are marked on it.
 - c. Fill in the latitudes and longitude in the given table. The first one is done for you.



Map of World

Sr. No	CITY	POSITION		LETTER
1	Beijing	39° N	116° E	
2	Washington			
3	Moscow			
4	Tokyo			
5	London			
6	Canberra			

1. Using a map scale, measure the distance between Karachi and Sukkur on a map of Sindh.
2. Using a map scale, measure the length of the river Indus on a map of Pakistan.
3. Buy a cheap plastic ball. Measure a point that is at an equal distance from the top and bottom of the ball, mark it as the Equator. Divide each hemisphere to show the remaining important lines of latitude.
4. Measure the length and breadth of your classroom. Now make a map of your classroom in your copy. Write a statement scale for it.

E. Extension activity

Turn on the index at the back of an atlas. We will see that every place shown in the atlas is listed in alphabetical order. We will find an entry like this: Multan 36 B3, 03.10 ° N 71.36° E. This tells us that the best place to find Multan is on page 36 in the grid B3. It also tells us that Multan is located at 30.10 ° N and 71.36° E. Turn to the index of your atlas and find the location of the three different cities.